信 息 工 程 学 院

《云计算》实验报告

学院：信息工程学院　**班级**：数科 2102　**姓名**：陈　**学号**:202101　**成绩**:

一、实验题目

MapReduce 编程

二、实验目的

掌握 MapReduce 编程。

理解 MapReduce 原理。

三、实验平台

希冀平台

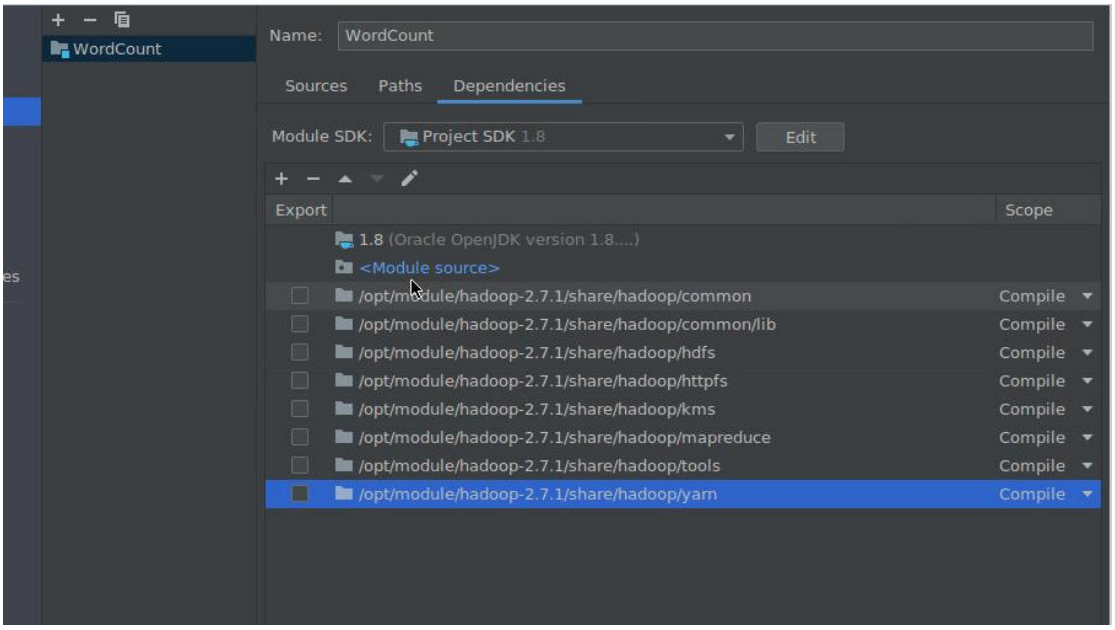
四、实验过程及步骤

实验 3

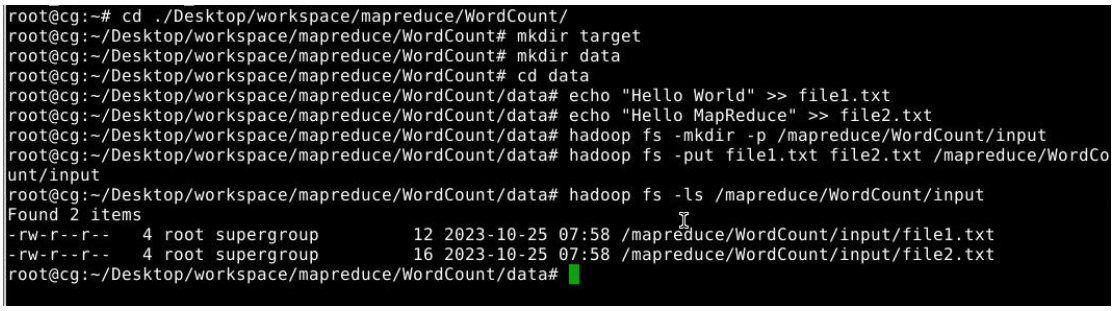
环境准备

```
master: starting namenode, logging to /opt/module/hadoop-2.7.1/logs/hadoop-root-
namenode-master.out
slave1: starting datanode, logging to /opt/module/hadoop-2.7.1/logs/hadoop-root-
datanode-slave1.out
slave2: starting datanode, logging to /opt/module/hadoop-2.7.1/logs/hadoop-root-
datanode-slave2.out
slave3: starting datanode, logging to /opt/module/hadoop-2.7.1/logs/hadoop-root-
datanode-slave3.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /opt/module/hadoop-2.7.1/logs/ha
doop-root-secondarynamenode-master.out
stopping yarn daemons
no resourcemanager to stop
slave2: no nodemanager to stop
slave3: no nodemanager to stop
slave1: no nodemanager to stop
no proxyserver to stop
starting yarn daemons
starting resourcemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root-res
ourcemanager-master.out
slave1: starting nodemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root
-nodemanager-slave1.out
slave3: starting nodemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root
-nodemanager-slave3.out
slave2: starting nodemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root
-nodemanager-slave2.out
no historyserver to stop
starting historyserver, logging to /opt/module/hadoop-2.7.1/logs/mapred-root-his
toryserver-master.out
root@master:~#
```

实验报告的内容与格式按任课教师的要求书写。

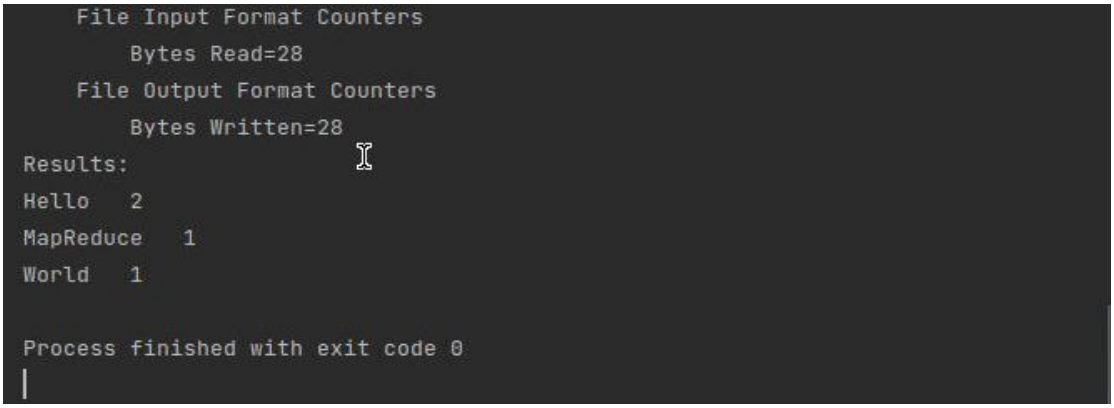


WordCount 测试数据准备



```
root@cg:~# cd ./Desktop/workspace/mapreduce/WordCount/
root@cg:~/Desktop/workspace/mapreduce/WordCount# mkdir target
root@cg:~/Desktop/workspace/mapreduce/WordCount# mkdir data
root@cg:~/Desktop/workspace/mapreduce/WordCount# cd data
root@cg:~/Desktop/workspace/mapreduce/WordCount/data# echo "Hello World" >> file1.txt
root@cg:~/Desktop/workspace/mapreduce/WordCount/data# echo "Hello MapReduce" >> file2.txt
root@cg:~/Desktop/workspace/mapreduce/WordCount/data# hadoop fs -mkdir -p /mapreduce/WordCount/input
root@cg:~/Desktop/workspace/mapreduce/WordCount/data# hadoop fs -put file1.txt file2.txt /mapreduce/WordCo
unt/input
root@cg:~/Desktop/workspace/mapreduce/WordCount/data# hadoop fs -ls /mapreduce/WordCount/input
Found 2 items
-rw-r--r--   4 root supergroup         12 2023-10-25 07:58 /mapreduce/WordCount/input/file1.txt
-rw-r--r--   4 root supergroup         16 2023-10-25 07:58 /mapreduce/WordCount/input/file2.txt
root@cg:~/Desktop/workspace/mapreduce/WordCount/data#
```

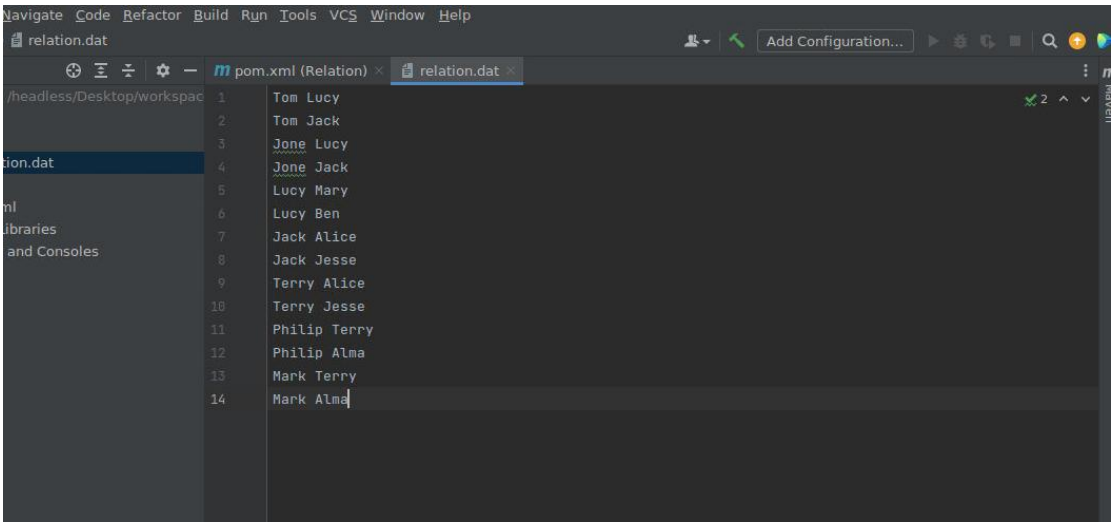
WordCount 运行结果

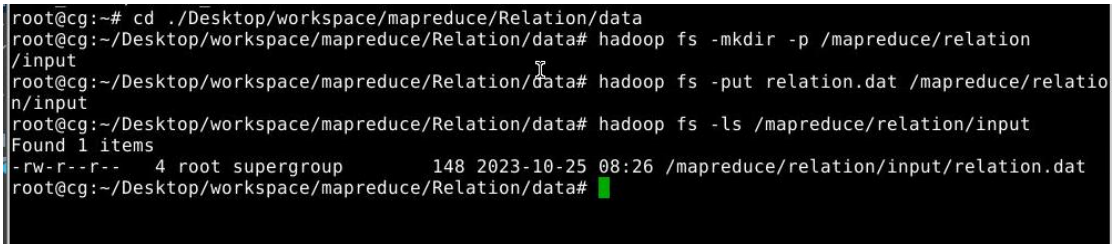


```
    File Input Format Counters
        Bytes Read=28
    File Output Format Counters
        Bytes Written=28
Results:
Hello   2
MapReduce   1
World   1

Process finished with exit code 0
```

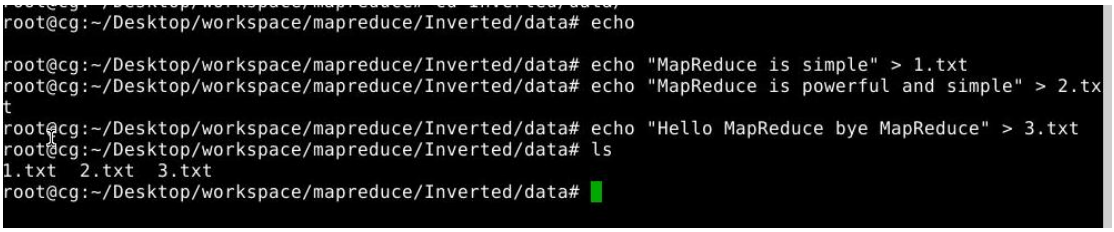
Relation 测试数据准备





Relation 实验结果

Inverted 测试数据准备



```
root@cg:~/Desktop/workspace/mapreduce/Inverted/data# hadoop fs -mkdir -p /mapreduce/inverted/input
root@cg:~/Desktop/workspace/mapreduce/Inverted/data# hadoop fs -put *.txt /mapreduce/inverted/input
root@cg:~/Desktop/workspace/mapreduce/Inverted/data# hadoop fs -ls /mapreduce/inverted/input
Found 3 items
-rw-r--r--   4 root supergroup         20 2023-10-25 08:38 /mapreduce/inverted/input/1.txt
-rw-r--r--   4 root supergroup         33 2023-10-25 08:38 /mapreduce/inverted/input/2.txt
-rw-r--r--   4 root supergroup         30 2023-10-25 08:38 /mapreduce/inverted/input/3.txt
root@cg:~/Desktop/workspace/mapreduce/Inverted/data#
```

Inverted 实验结果

```
Results:
Hello   3.txt:1 ;
MapReduce   3.txt:2 ; 1.txt:1 ; 2.txt:1 ;
and 2.txt:1 ;
bye 3.txt:1 ;
is  1.txt:1 ; 2.txt:1 ;
powerful    2.txt:1 ;
simple  2.txt:1 ; 1.txt:1 ;

Process finished with exit code 0
```

PageRank 测试数据准备

```
root@cg:~/Desktop/workspace/mapreduce/PageRank/data# hadoop fs -mkdir /mapreduce/PageRank/input
mkdir: `/mapreduce/PageRank/input': No such file or directory
Error: Java Execution Failed.
root@cg:~/Desktop/workspace/mapreduce/PageRank/data# hadoop fs -mkdir -p /mapreduce/PageRank/input
root@cg:~/Desktop/workspace/mapreduce/PageRank/data# hadoop fs -put input.dat /mapreduce/PageRank/
nput
root@cg:~/Desktop/workspace/mapreduce/PageRank/data#
```

PageRank 实验结果

```
Results:
A   0.10800889 C D
B   0.13810074 A D
C   0.6157898 C
D   0.13810074 B C
```

```
drwxr-xr-x   - root supergroup          0 2023-10-25 08:53 /mapreduce/PageRank/input
drwxr-xr-x   - root supergroup          0 2023-10-25 08:57 /mapreduce/PageRank/output
drwxr-xr-x   - root supergroup          0 2023-10-25 08:57 /mapreduce/PageRank/output1
drwxr-xr-x   - root supergroup          0 2023-10-25 08:57 /mapreduce/PageRank/output12
drwxr-xr-x   - root supergroup          0 2023-10-25 08:57 /mapreduce/PageRank/output123
drwxr-xr-x   - root supergroup          0 2023-10-25 08:57 /mapreduce/PageRank/output1234
root@cg:~/Desktop/workspace/mapreduce/PageRank/data#
```

流量统计

测试数据准备

```
USER_ID: 0, GROUP_ID: 0
root@cg:~# cd ./Desktop/workspace/mapreduce/Test1/
root@cg:~/Desktop/workspace/mapreduce/Test1# hadoop fs -mkdir -p /mapreduce/Test
1/input
root@cg:~/Desktop/workspace/mapreduce/Test1# hadoop fs -put data.dat /mapreduce/
Test1/input
root@cg:~/Desktop/workspace/mapreduce/Test1# hadoop fs -ls /mapreduce/Test1/inpu
t
Found 1 items
-rw-r--r--   4 root supergroup       1285 2023-10-29 08:27 /mapreduce/Test1/inpu
t/data.dat
root@cg:~/Desktop/workspace/mapreduce/Test1#
```

运行结果

```
    File Output Format Counters
        Bytes Written=159
Results:
13726230503 73386
13726230513 248
13726230533 27162
13726230543 3633
13826230523 54324
13826230563 27162
13926230553 27162
13926230573 27162
18912688533 29502

Process finished with exit code 0
```

代码：

MyMapper：

```
import  java.io.IOException;

import  org.apache.commons.lang.StringUtils;

import  org.apache.hadoop.io.LongWritable;

import  org.apache.hadoop.io.Text;

import  org.apache.hadoop.mapreduce.Mapper;
```

```
public class MyMapper extends Mapper<LongWritable, Text, Text, LongWri
table>{

    @Override

    protected void map(LongWritable key, Text value, Context context)

            throws IOException, InterruptedException {

        String str = value.toString();

        String[] mes = StringUtils.split(str," ");

        int i=0;

        for(String tmp : mes){

            if(i == mes.length-2 || i == mes.length-3){

                context.write(new Text(mes[0]), new LongWritable(Integer.
parseInt(tmp)));

            }

            i++;

        }

    }

}
```

MyReducer：

```
import java.io.IOException;

import org.apache.hadoop.io.LongWritable;

import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Reducer;


public class MyReducer extends Reducer<Text, LongWritable, Text, LongWr
itable>{
```

```
    @Override
    protected void reduce(Text key, Iterable<LongWritable> values,Context context)
            throws IOException, InterruptedException {
        long cnt = 0;
        for(LongWritable value : values){
            cnt += value.get();
        }
        context.write(key, new LongWritable(cnt));
    }
}
```

MyRunner：

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import java.util.Scanner;
import org.apache.hadoop.fs.FSDataInputStream;
import org.apache.hadoop.fs.FileSystem;
import java.net.URI;
```

```
public class MyRunner {

    public static void main(String[] args) throws IOException, ClassNotFou
ndException, InterruptedException {

        Configuration conf = new Configuration();

        Job job = Job.getInstance(conf);



        job.setJarByClass(MyRunner.class);



        job.setMapperClass(MyMapper.class);

        job.setReducerClass(MyReducer.class);



        job.setOutputKeyClass(Text.class);

        job.setOutputValueClass(Text.class);




        job.setMapOutputKeyClass(Text.class);

        job.setMapOutputValueClass(LongWritable.class);



        Scanner sc = new Scanner(System.in);

        System.out.print("inputPath:");

        String inputPath = sc.next();

        System.out.print("outputPath:");

        String outputPath = sc.next();
```

```
        FileInputFormat.setInputPaths(job, new Path("hdfs://master:9000"+in
putPath));


        FileOutputFormat.setOutputPath(job, new Path("hdfs://master:9000"+
outputPath));


        job.waitForCompletion(true);


        try {
            FileSystem fs = FileSystem.get(new URI("hdfs://master:9000"),
 new Configuration());
            Path srcPath = new Path(outputPath+"/part-r-00000");


            FSDataInputStream is = fs.open(srcPath);
            System.out.println("Results:");
            while(true) {
                String line = is.readLine();
                if(line == null) {
                    break;
                }
                System.out.println(line);
            }
            is.close();
        }catch(Exception e) {
```

```
                e.printStackTrace();
            }
        }
    }
```

实验 4

开启服务

```
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /opt/module/hadoop-2.7.1/logs/ha
doop-root-secondarynamenode-master.out
stopping yarn daemons
no resourcemanager to stop
slave2: no nodemanager to stop
slave1: no nodemanager to stop
slave3: no nodemanager to stop
no proxyserver to stop
starting yarn daemons
starting resourcemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root-res
ourcemanager-master.out
slave1: starting nodemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root
-nodemanager-slave1.out
slave2: starting nodemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root
-nodemanager-slave2.out
slave3: starting nodemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root
-nodemanager-slave3.out
no historyserver to stop
starting historyserver, logging to /opt/module/hadoop-2.7.1/logs/mapred-root-his
toryserver-master.out
root@master:~#
```

数据准备

```
root@cg:~# ls
Desktop    install  Pictures                   Templates  wm.log
Documents  Music    Public                     users.txt  wm_startup.sh
Downloads  noVNC    rstudio-1.0.136-amd64.deb  Videos
root@cg:~# vim login_logs.txt
root@cg:~# ls
Desktop    install         noVNC     rstudio-1.0.136-amd64.deb  Videos
Documents  login_logs.txt  Pictures  Templates                  wm.log
Downloads  Music           Public    users.txt                  wm_startup.sh
root@cg:~# hdfs dfs -mkdir -p /join/in
root@cg:~# hdfs dfs -mkdir -p /join/out
root@cg:~# hdfs dfs -put users.txt /join/in
root@cg:~# hdfs dfs -put login_logs.txt /join/in
root@cg:~#
```

运行结果 1

```
        WRONG_MAP=0
        WRONG_REDUCE=0
    File Input Format Counters
        Bytes Read=238
    File Output Format Counters
        Bytes Written=241

 Process finished with exit code 0
```

```
root@cg:~# hdfs dfs -put login_logs.txt /join/in
root@cg:~# hdfs dfs -cat /join/out/*
1       LiXiaolong      2015-06-07 15:26:11     192.168.137.103
1       LiXiaolong      2015-06-07 15:22:38     192.168.137.101
1       LiXiaolong      2015-06-07 15:10:18     192.168.137.101
3       Zhangsan        2015-06-07 15:18:36     192.168.137.102
3       Zhangsan        2015-06-07 15:12:18     192.168.137.102
root@cg:~#
```

运行结果 2

```
Run:  DistributedCacheDemo
        WRONG_MAP=0
        WRONG_REDUCE=0
    File Input Format Counters
        Bytes Read=190
    File Output Format Counters
        Bytes Written=241

 Process finished with exit code 0
```

```
3       Zhangsan        2015-06-07 15:12:18     192.168.137.102
root@cg:~# hdfs dfs -cat /join/out/*
1       LiXiaolong      2015-06-07 15:26:11     192.168.137.103
1       LiXiaolong      2015-06-07 15:22:38     192.168.137.101
1       LiXiaolong      2015-06-07 15:10:18     192.168.137.101
3       Zhangsan        2015-06-07 15:18:36     192.168.137.102
3       Zhangsan        2015-06-07 15:12:18     192.168.137.102
root@cg:~#
```

启动 mysql 服务

```
root@master:~# /etc/init.d/mysql start
 * Starting MySQL database server mysqld
su: warning: cannot change directory to /nonexistent: No such file or directory
                                                                        [ OK ]
root@master:~#
```

创建目录

```
root@cg:~# hdfs dfs -mkdir -p /join/in/users
root@cg:~# hdfs dfs -mkdir -p /join/in/login_logs
root@cg:~# hdfs dfs -mv /join/in/users.txt /join/in/users
root@cg:~# hdfs dfs -mv /join/in/login_logs.txt /join/in/login_logs
root@cg:~# hive
```

创建外部表

```
hive> create external table users(userid int, name string) row format delimited
fields terminated by '\t' location '/join/in/users';
OK
Time taken: 0.419 seconds
hive>
```

```
hive> create external table login_logs(userid string,login_time string,login_ip
string) row format delimited fields terminated by '\t' location '/join/in/login_
logs';
OK
Time taken: 0.085 seconds
hive>
```

执行连接查询并保持结果

```
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
Execution log at: /tmp/root/root_20231029091640_196516b6-2b4a-4f97-a7fe-8f40e47c
7ee9.log
2023-10-29 09:16:43     Starting to launch local task to process map join;     m
aximum memory = 477626368
2023-10-29 09:16:44     Dump the side-table for tag: 0 with group count: 5 into
file: file:/tmp/root/86c25e2a-9fbe-402b-934f-23d2a86bb801/hive_2023-10-29_09-16-
40_562_7772238984785708143-1/-local-10003/HashTable-Stage-4/MapJoin-mapfile00--.
hashtable
2023-10-29 09:16:44     Uploaded 1 File to: file:/tmp/root/86c25e2a-9fbe-402b-93
4f-23d2a86bb801/hive_2023-10-29_09-16-40_562_7772238984785708143-1/-local-10003/
HashTable-Stage-4/MapJoin-mapfile00--.hashtable (433 bytes)
2023-10-29 09:16:44     End of local task; Time Taken: 0.955 sec.
Execution completed successfully
MapredLocal task succeeded
Launching Job 1 out of 1
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1698570301613_0001, Tracking URL = http://master:8088/proxy/a
pplication_1698570301613_0001/
Kill Command = /opt/module/hadoop-2.7.1/bin/hadoop job  -kill job_1698570301613_
0001
Hadoop job information for Stage-4: number of mappers: 1; number of reducers: 0
2023-10-29 09:16:51,630 Stage-4 map = 0%,  reduce = 0%
2023-10-29 09:16:57,954 Stage-4 map = 100%,  reduce = 0%, Cumulative CPU 3.57 se
c
MapReduce Total cumulative CPU time: 3 seconds 570 msec
Ended Job = job_1698570301613_0001
Moving data to: hdfs://master:9000/user/hive/warehouse/user_login_logs
Table default.user_login_logs stats: [numFiles=1, numRows=5, totalSize=241, rawD
ataSize=236]
MapReduce Jobs Launched:
Stage-Stage-4: Map: 1   Cumulative CPU: 3.57 sec   HDFS Read: 6882 HDFS Write: 3
21 SUCCESS
Total MapReduce CPU Time Spent: 3 seconds 570 msec
OK
Time taken: 18.76 seconds
hive>
```

查看结果

```
hive> select * from user_login_logs;
OK
1       LiXiaolong      2015-06-07 15:10:18     192.168.137.101
3       Zhangsan        2015-06-07 15:12:18     192.168.137.102
3       Zhangsan        2015-06-07 15:18:36     192.168.137.102
1       LiXiaolong      2015-06-07 15:22:38     192.168.137.101
1       LiXiaolong      2015-06-07 15:26:11     192.168.137.103
Time taken: 0.095 seconds, Fetched: 5 row(s)
hive>
```

五、实验总结

通过本次实验我学习了 MapReduce 编程，并且针对一些简单的问题进行处理，进一步了解了 Map 阶段对任务进行分解，Reduce 阶段对任务进行合并的过程。也学习到了如何通过编写 MapReduce 程序来完成 join 操作。并知道了在执行效率上 map-join>SemiJoin>reduce-join。